10.18 Consider the data file *mroz* on working wives. Use the 428 observations on married women who participate in the labor force. In this exercise, we examine the effectiveness of a parent's college education as an instrumental variable.

a.

a. Create two new variables. *MOTHERCOLL* is a dummy variable equaling one if *MOTHEREDUC* > 12, zero otherwise. Similarly, *FATHERCOLL* equals one if *FATHEREDUC* > 12 and zero otherwise. What percentage of parents have some college education in this sample?

```
> library(POE5Rdata)
> library(AER)  # for ivreg
> data(mroz)
>
> mroz1 <- mroz[mroz$lfp == 1, ]
> nrow(mroz1)
[1] 428
>
> # -- Part a ----------------------------------
> mroz1$MOTHERCOLL <- as.integer(mroz1$mothereduc > 12)
> mroz1$FATHERCOLL <- as.integer(mroz1$fathereduc > 12)
>
> mean(mroz1$MOTHERCOLL)
[1] 0.1214953
> mean(mroz1$FATHERCOLL)
[1] 0.1168224
> mean(mroz1$MOTHERCOLL == 1 | mroz1$FATHERCOLL == 1)
[1] 0.1869159
```

⇒ 12.14953 % of mothers have some college education in this sample.
11.68224 % of fathers have some college education in this sample.
18.69159 % of parents (mother or father) have some college education in this sample.

b.

b. Find the correlations between *EDUC*, *MOTHERCOLL*, and *FATHERCOLL*. Are the magnitudes of these correlations important? Can you make a logical argument why *MOTHERCOLL* and *FATHERCOLL* might be better instruments than *MOTHEREDUC* and *FATHEREDUC*?

```
> # -- Part b ---------------------------
> cor(mroz1[, c("educ", "MOTHERCOLL", "FATHERCOLL")])
                educ MOTHERCOLL FATHERCOLL
educ       1.0000000  0.3594705  0.3984962
MOTHERCOLL 0.3594705  1.0000000  0.3545709
FATHERCOLL 0.3984962  0.3545709  1.0000000
```

The magnitudes of these correlations are important (>0.2).
One logical argument is that MOTHEREDUC and FATHEREDUC may

directly affects the WAGE, while MOTHERCOLL and FATHERCOLL can only equal to one or zero, so the information they contain may not be enough, making them less likely to directly affect the WAGE, which is one of the condition of a good IV.

C.

c. Estimate the wage equation in Example 10.5 using *MOTHERCOLL* as the instrumental variable. What is the 95% interval estimate for the coefficient of *EDUC*?

```
> # —— Part c : IV: MOTHERCOLL ————————————————————————————————————
> iv_c <- ivreg(log(wage) ~ exper + I(exper^2) + educ |
+                 exper + I(exper^2) + MOTHERCOLL,
+               data = mroz1)
> summary(iv_c)

Call:
ivreg(formula = log(wage) ~ exper + I(exper^2) + educ | exper +
    I(exper^2) + MOTHERCOLL, data = mroz1)

Residuals:
     Min       1Q   Median       3Q      Max
-3.08719 -0.32444  0.04147  0.36634  2.35621

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.1327561  0.4965325  -0.267  0.78932
exper        0.0433444  0.0134135   3.231  0.00133 **
I(exper^2)  -0.0008711  0.0004017  -2.169  0.03066 *
educ         0.0760180  0.0394077   1.929  0.05440 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6703 on 424 degrees of freedom
Multiple R-Squared: 0.147,   Adjusted R-squared: 0.1409
Wald test:   8.2 on 3 and 424 DF,  p-value: 2.569e-05

> confint(iv_c, "educ", level = 0.95)
            2.5 %    97.5 %
educ -0.001219763 0.1532557
```

95% interval: [-0.001219763, 0.1532557].

d.

d. For the problem in part (c), estimate the first-stage equation. What is the value of the *F*-test statistic for the hypothesis that *MOTHERCOLL* has no effect on *EDUC*? Is *MOTHERCOLL* a strong instrument?

```
> # -- Part d: First stage equation ---------------------------------
> library(car)
> first_d <- lm(educ ~ exper + I(exper^2) + MOTHERCOLL, data = mroz1)
> summary(first_d)

Call:
lm(formula = educ ~ exper + I(exper^2) + MOTHERCOLL, data = mroz1)

Residuals:
    Min      1Q  Median      3Q     Max 
-7.4267 -0.4826 -0.3731  1.0000  4.9353 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 12.079094   0.303118  39.849  < 2e-16 ***
exper        0.056230   0.042101   1.336    0.182    
I(exper^2)  -0.001956   0.001256  -1.557    0.120    
MOTHERCOLL   2.517068   0.315713   7.973 1.46e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.133 on 424 degrees of freedom
Multiple R-squared:  0.1347, Adjusted R-squared:  0.1285 
F-statistic: 21.99 on 3 and 424 DF,  p-value: 2.965e-13

> linearHypothesis(first_d, "MOTHERCOLL = 0")

Linear hypothesis test:
MOTHERCOLL = 0

Model 1: restricted model
Model 2: educ ~ exper + I(exper^2) + MOTHERCOLL

  Res.Df    RSS Df Sum of Sq      F    Pr(>F)    
1    425 2219.2                                  
2    424 1929.9  1    289.32 63.563 1.455e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

F-test statistics : 63.563

Hence MOTHERCOLL is a strong instrument.

e.

e. Estimate the wage equation in Example 10.5 using *MOTHERCOLL* and *FATHERCOLL* as the instrumental variables. What is the 95% interval estimate for the coefficient of *EDUC*? Is it narrower or wider than the one in part (c)?

```
> # —— Part e: IV: MOTHERCOLL + FATHERCOLL ——————————————————————
> iv_e <- ivreg(log(wage) ~ exper + I(exper^2) + educ |
+                 exper + I(exper^2) + MOTHERCOLL + FATHERCOLL,
+               data = mroz1)
> summary(iv_e)

Call:
ivreg(formula = log(wage) ~ exper + I(exper^2) + educ | exper +
    I(exper^2) + MOTHERCOLL + FATHERCOLL, data = mroz1)

Residuals:
     Min       1Q   Median       3Q      Max
-3.07797 -0.32128  0.03418  0.37648  2.36183

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.2790819  0.3922213  -0.712  0.47714
exper        0.0426761  0.0132950   3.210  0.00143 **
I(exper^2)  -0.0008486  0.0003976  -2.135  0.03337 *
educ         0.0878477  0.0307808   2.854  0.00453 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6679 on 424 degrees of freedom
Multiple R-Squared: 0.153,   Adjusted R-squared: 0.147
Wald test: 9.724 on 3 and 424 DF,  p-value: 3.224e-06

> confint(iv_e, "educ", level = 0.95)
          2.5 %    97.5 %
educ 0.02751845 0.1481769
```

95% interval: [0.02751845, 0.1481769],

which is narrower than the one in part (c).

f、

f. For the problem in part (e), estimate the first-stage equation. Test the joint significance of *MOTHERCOLL* and *FATHERCOLL*. Do these instruments seem adequately strong?

```
> # —— Part f: First stage equation ——————————————————————————————————
> first_f <- lm(educ ~ exper + I(exper^2) + MOTHERCOLL + FATHERCOLL, data = mroz1)
> summary(first_f)

Call:
lm(formula = educ ~ exper + I(exper^2) + MOTHERCOLL + FATHERCOLL,
    data = mroz1)

Residuals:
    Min      1Q  Median      3Q     Max
-7.2152 -0.3056 -0.2152  0.7627  5.0620

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 11.890259   0.290251  40.965  < 2e-16 ***
exper        0.049149   0.040133   1.225    0.221
I(exper^2)  -0.001449   0.001199  -1.209    0.227
MOTHERCOLL   1.749947   0.322347   5.429 9.58e-08 ***
FATHERCOLL   2.186612   0.329917   6.628 1.04e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.033 on 423 degrees of freedom
Multiple R-squared:  0.2161, Adjusted R-squared:  0.2086
F-statistic: 29.15 on 4 and 423 DF,  p-value: < 2.2e-16

> linearHypothesis(first_f, c("MOTHERCOLL = 0", "FATHERCOLL = 0"))

Linear hypothesis test:
MOTHERCOLL = 0
FATHERCOLL = 0

Model 1: restricted model
Model 2: educ ~ exper + I(exper^2) + MOTHERCOLL + FATHERCOLL

  Res.Df    RSS Df Sum of Sq      F    Pr(>F)
1    425 2219.2
2    423 1748.3  2    470.88 56.963 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The p-value is 2.2×10^{-16},
which shows that these instruments seem adequately strong.

g.

g. For the IV estimation in part (e), test the validity of the surplus instrument. What do you conclude?

```
> # —— Part g: Sargan test ——————————————————————————————————————————
> mroz1$ehat <- residuals(iv_e)
> sargan_reg <- lm(ehat ~ exper + I(exper^2) + MOTHERCOLL + FATHERCOLL, data = mroz1)
> n <- nrow(mroz1)
> R2 <- summary(sargan_reg)$r.squared
> sargan_stat <- n * R2
> sargan_stat
[1] 0.2375851
> p_value <- 1 - pchisq(sargan_stat, df = 1)  # df = 超額工具數 = 2-1 = 1
> p_value
[1] 0.6259557
```

The p-value is large, which shows that we can't reject the validity of the surplus instrument. So we accept that these instruments are valid.

10.20 The CAPM [see Exercises 10.14 and 2.16] says that the risk premium on security j is related to the risk premium on the market portfolio. That is

$$r_j - r_f = \alpha_j + \beta_j(r_m - r_f)$$

where r_j and r_f are the returns to security j and the risk-free rate, respectively, r_m is the return on the market portfolio, and β_j is the jth security's "*beta*" value. We measure the market portfolio using the Standard & Poor's value weighted index, and the risk-free rate by the 30-day LIBOR monthly rate of return. As noted in Exercise 10.14, if the market return is measured with error, then we face an errors-in-variables, or measurement error, problem.

a. Use the observations on Microsoft in the data file *capm5* to estimate the CAPM model using OLS. How would you classify the Microsoft stock over this period? Risky or relatively safe, relative to the market portfolio?

a.

```
> library(POE5Rdata)
> library(AER)
> library(car)
> data(capm5)
> ?capm5
>
> capm5$excess_msft <- capm5$msft - capm5$riskfree   # r_j - r_f
> capm5$excess_mkt  <- capm5$mkt  - capm5$riskfree   # r_m - r_f
>
> # —— Part a：OLS 估計 CAPM ——————————————————————————————————————————
> ols_a <- lm(excess_msft ~ excess_mkt, data = capm5)
> summary(ols_a)

Call:
lm(formula = excess_msft ~ excess_mkt, data = capm5)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.27424 -0.04744 -0.00820  0.03869  0.35801 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.003250   0.006036   0.538    0.591    
excess_mkt  1.201840   0.122152   9.839   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08083 on 178 degrees of freedom
Multiple R-squared:  0.3523, Adjusted R-squared:  0.3486 
F-statistic:  96.8 on 1 and 178 DF,  p-value: < 2.2e-16
```

The estimated beta is 1.201840, which is larger than 1. This shows that Microsoft is risky relative to market portfolio.

b.

b. It has been suggested that it is possible to construct an IV by ranking the values of the explanatory variable and using the rank as the IV, that is, we sort $(r_m - r_f)$ from smallest to largest, and assign the values $RANK = 1, 2, \ldots, 180$. Does this variable potentially satisfy the conditions IV1–IV3? Create *RANK* and obtain the first-stage regression results. Is the coefficient of *RANK* very significant? What is the R^2 of the first-stage regression? Can *RANK* be regarded as a strong IV?

```
> # —— Part b：建立 RANK 工具變數 ————————————————————————————————————————
> capm5$RANK <- rank(capm5$excess_mkt)   # 從最小到最大排序，賦值 1~180
>
> first_b <- lm(excess_mkt ~ RANK, data = capm5)
> summary(first_b)

Call:
lm(formula = excess_mkt ~ RANK, data = capm5)

Residuals:
      Min        1Q    Median        3Q       Max
-0.110497 -0.006308  0.001497  0.009433  0.029513

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -7.903e-02  2.195e-03   -36.0   <2e-16 ***
RANK         9.067e-04  2.104e-05    43.1   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01467 on 178 degrees of freedom
Multiple R-squared:  0.9126, Adjusted R-squared:  0.9121
F-statistic:  1858 on 1 and 178 DF,  p-value: < 2.2e-16
```

IV1: Yes, it's likely that the RANK doesn't have a direct affect on $r_j - r_f$

IV2: Yes, it's likely that the RANK is exogenous.

IV3: Yes, the RANK is indeed correlated with $r_m - r_f$.

So it does have potential to satisfy the conditions IV1 ~ IV3.

The coefficient of RANK does very significant ($p\text{-value} < 2 \times 10^{-16}$)

The R^2 is 0.9126, which is very high. So RANK can be regarded as a strong IV.

C.

c. Compute the first-stage residuals, $\hat{v}$, and add them to the CAPM model. Estimate the resulting augmented equation by OLS and test the significance of $\hat{v}$ at the 1% level of significance. Can we conclude that the market return is exogenous?

```
> # —— Part c：Hausman 檢定 (用 RANK 的第一階段殘差) ————————————————————————
> capm5$vhat_c <- residuals(first_b)
>
> # 將殘差加入原始方程式，OLS 估計擴充方程式
> hausman_c <- lm(excess_msft ~ excess_mkt + vhat_c, data = capm5)
> summary(hausman_c)

Call:
lm(formula = excess_msft ~ excess_mkt + vhat_c, data = capm5)

Residuals:
     Min       1Q   Median       3Q      Max
-0.27140 -0.04213 -0.00911  0.03423  0.34887

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003018   0.005984   0.504   0.6146
excess_mkt   1.278318   0.126749  10.085   <2e-16 ***
vhat_c      -0.874599   0.428626  -2.040   0.0428 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08012 on 177 degrees of freedom
Multiple R-squared:  0.3672, Adjusted R-squared:   0.36
F-statistic: 51.34 on 2 and 177 DF,  p-value: < 2.2e-16
```

The p-value of $\hat{v}$ is only 0.0428, so it is not significant at the 1% level of significance.
Thus we can't reject that the market return is exogenous.
So we accept that the market return is exogenous.

d.

d. Use *RANK* as an IV and estimate the CAPM model by IV/2SLS. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?

```
> # —— Part d：IV/2SLS，工具變數 = RANK ————————————————————————————————————
> iv_d <- ivreg(excess_msft ~ excess_mkt | RANK, data = capm5)
> summary(iv_d)

Call:
ivreg(formula = excess_msft ~ excess_mkt | RANK, data = capm5)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.271625 -0.049675 -0.009693  0.037683  0.355579 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.003018   0.006044   0.499    0.618    
excess_mkt  1.278318   0.128011   9.986   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08092 on 178 degrees of freedom
Multiple R-Squared: 0.3508,  Adjusted R-squared: 0.3472 
Wald test: 99.72 on 1 and 178 DF,  p-value: < 2.2e-16
```

	Estimate	Std. Error
OLS (part a)	1.201840	0.122152
IV/2SLS (part d)	1.278318	0.128011

Note that the std. error in IV/2SLS is larger than that in OLS. This agrees with my expectation since the information of RANK is less than the variable excess_mkt. So it is not surprised that the std. error is larger in the IV/2SLS method.

e.

e. Create a new variable $POS = 1$ if the market return $(r_m - r_f)$ is positive, and zero otherwise. Obtain the first-stage regression results using both $RANK$ and POS as instrumental variables. Test the joint significance of the IV. Can we conclude that we have adequately strong IV? What is the R^2 of the first-stage regression?

```
> # — Part e：建立 POS，第一階段加入 RANK + POS ——————————————————————————
> capm5$POS <- as.integer(capm5$excess_mkt > 0)   # 市場超額報酬為正 → 1
>
> first_e <- lm(excess_mkt ~ RANK + POS, data = capm5)
> summary(first_e)

Call:
lm(formula = excess_mkt ~ RANK + POS, data = capm5)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.109182 -0.006732  0.002858  0.008936  0.026652 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.0804216  0.0022622  -35.55   <2e-16 ***
RANK         0.0009819  0.0000400   24.55   <2e-16 ***
POS         -0.0092762  0.0042156   -2.20   0.0291 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01451 on 177 degrees of freedom
Multiple R-squared:  0.9149, Adjusted R-squared:  0.9139 
F-statistic: 951.3 on 2 and 177 DF,  p-value: < 2.2e-16

> # 聯合顯著性 F-test
> linearHypothesis(first_e, c("RANK = 0", "POS = 0"))

Linear hypothesis test:
RANK = 0
POS = 0

Model 1: restricted model
Model 2: excess_mkt ~ RANK + POS

  Res.Df     RSS Df Sum of Sq      F    Pr(>F)    
1    179 0.43784                                  
2    177 0.03727  2   0.40057 951.26 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

We can see that the R^2 of the first-stage regression is 0.9149, which is higher than the R^2 in (b). Also, the F-test statistics is 951.26, which is very high. So we can conclude that we have adequately strong IV.

f.

f. Carry out the Hausman test for endogeneity using the residuals from the first-stage equation in (e). Can we conclude that the market return is exogenous at the 1% level of significance?

```
> # —— Part f：Hausman 檢定（用 part e 的第一階段殘差） ————————————————————
> capm5$vhat_e <- residuals(first_e)
>
> hausman_f <- lm(excess_msft ~ excess_mkt + vhat_e, data = capm5)
> summary(hausman_f)

Call:
lm(formula = excess_msft ~ excess_mkt + vhat_e, data = capm5)

Residuals:
     Min       1Q   Median       3Q      Max
-0.27132 -0.04261 -0.00812  0.03343  0.34867

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003004   0.005972   0.503   0.6157
excess_mkt   1.283118   0.126344  10.156   <2e-16 ***
vhat_e      -0.954918   0.433062  -2.205   0.0287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07996 on 177 degrees of freedom
Multiple R-squared:  0.3696, Adjusted R-squared:  0.3625
F-statistic: 51.88 on 2 and 177 DF,  p-value: < 2.2e-16
```

The p-value of $\hat{v}$ is only 0.0287, so it is not significant at the 1% level of significance.
Thus we can't reject that the market return is exogenous.
So we accept that the market return is exogenous.

g.

g. Obtain the IV/2SLS estimates of the CAPM model using *RANK* and *POS* as instrumental variables. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?

```
> # —— Part g：IV/2SLS，工具變數 = RANK + POS ——————————————————————————————
> iv_g <- ivreg(excess_msft ~ excess_mkt | RANK + POS, data = capm5)
> summary(iv_g)

Call:
ivreg(formula = excess_msft ~ excess_mkt | RANK + POS, data = capm5)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.27168 -0.04960 -0.00983  0.03762  0.35543 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.003004   0.006044   0.497     0.62    
excess_mkt  1.283118   0.127866  10.035   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08093 on 178 degrees of freedom
Multiple R-Squared: 0.3507,  Adjusted R-squared: 0.347 
Wald test: 100.7 on 1 and 178 DF,  p-value: < 2.2e-16
```

	Estimate	Std. Error
OLS (part a)	1.201840	0.122152
IV/2SLS (part g)	1.283118	0.127866

Note that the std. error in IV/2SLS is larger than that in OLS. This agrees with my expectation since the information of RANK and POS is less than the variable excess_mkt. So it is not surprised that the std. error is larger in the IV/2SLS method.

h.

h. Obtain the IV/2SLS residuals from part (g) and use them (not an automatic command) to carry out a Sargan test for the validity of the surplus IV at the 5% level of significance.

```
> # —— Part h：手動 Sargan 檢定 (5% 顯著水準) ————————————————————————————
> # 超額識別：2 個工具 - 1 個內生變數 = 1 個自由度
> capm5$ehat_g <- residuals(iv_g)
>
> sargan_reg <- lm(ehat_g ~ RANK + POS, data = capm5)
> n  <- nrow(capm5)
> R2 <- summary(sargan_reg)$r.squared
> sargan_stat <- n * R2
> sargan_stat
[1] 0.5584634
> p_value <- 1 - pchisq(sargan_stat, df = 1)   # df = 2 - 1 = 1
> p_value
[1] 0.45488
> # p > 0.05 → 無法拒絕工具有效性，surplus IV (POS) 通過 Sargan 檢定
```

Since $p > 0.05$, we can't reject the validity of the surplus IV at the 5% level of significance.